1-1-25

Continuity :- (Lec # 05)

- Continuity depends on limit.

A function $f(x,y)$ is continuous at (a,b) If $\lim_{(x,y)\to(a,b)} f(x,y) = f(a,b)$.

We say that $f(x,y)$ is continuous on D (domain) if $f(x,y)$ is continuous at every point (a,b) in D

Exp :- where is $f(x,y) = \frac{x^2 - y^2}{x^2 + y^2}$ continuous?

Solution :-

funct is discontinuous at origin where $x = 0$, $y = 0$.

OR

$f(x,y)$ is discontinuous on $(0,0)$ and will continuous on all real numbers except this point.

next

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B/c it is not defined there.

Since $f(x,y)$ is a rational fn; it is continuous on its domain, which is set
$D = \{(x,y) / (x,y) \neq (0,0)\}$.

In short, fn (x,y) is continuous on and $(x,y) \in \mathbb{R}^2$ except at origin $(0,0)$.

Exp:-

is $g(x,y) = \begin{cases} \frac{x^2-y^2}{x^2+y^2}, & \text{If } (x,y) \neq (0,0) \\ 0, & \text{If } (x,y) = (0,0) \end{cases}$

Solution:-

~~continuous~~

It is peace wise function.
Peace wise = function in pieces.

$g(x,y) = \frac{x^2-y^2}{x^2+y^2}$ is discontinuous at $(0,0)$.

$g(x,y) = 0$, is continuous at $(0,0)$.

Now check the limit either exist or not?

at x-axis, $y=0$	at y-axis, $x=0$
$\lim_{(x \to 0)} \frac{x^2-0^2}{x^2+0^2}$	$\lim_{(y \to 0)} \frac{0^2-y^2}{0^2+y^2}$
$= \frac{\cancel{x^2}}{\cancel{x^2}} = 1$	$= \frac{-\cancel{y^2}}{\cancel{y^2}} = -1$

limit doesn't exist b/c of different lies.

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Since, f has two different limits along two different lines, therefore limit of $g(x,y)$ doesn't exist, therefore the given function $g(x,y)$ is discontinuous.

Exp:-

is $f(x,y) = \begin{cases} \dfrac{3x^2y}{x^2+y^2}, & \text{if } (x,y) \neq (0,0) \\ 0, & \text{if } (x,y) = (0,0). \end{cases}$

Solution:-

Peace wise function.

$g(x,y) = \dfrac{3x^2y}{x^2+y^2}$ is discontinuous at $(0,0)$

$g(x,y) = 0$ is continuous at $(0,0)$.

Check, limit exists or not.

at x. axis, $y = 0$

$$\lim_{(x\to 0)} \frac{3x^2(0)}{x^2+(0)^2}$$

$$= \frac{0}{x^2} = 0$$

at y. axis, $x = 0$

$$\lim_{(y\to 0)} \frac{3(0)^2 y}{(0)^2+y^2}$$

$$= \frac{0}{y^2} = 0$$

Third line $y = x$

$$\lim_{(x,y)\to(0,0)} \frac{3x^2y}{x^2+y^2}$$

$$= \frac{3x^2(x)}{x^2+(x)^2} = \frac{3x^3}{2x^2} = \boxed{\frac{3x}{2}}$$

$$= \frac{3x}{2} = \frac{0}{2} = 0$$

$\boxed{\text{limit exists}}$